Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. We trained each model for 20 epochs with a batch size of 64.

2 Results

Figure 1 shows the validation accuracy of Baseline by dataset.

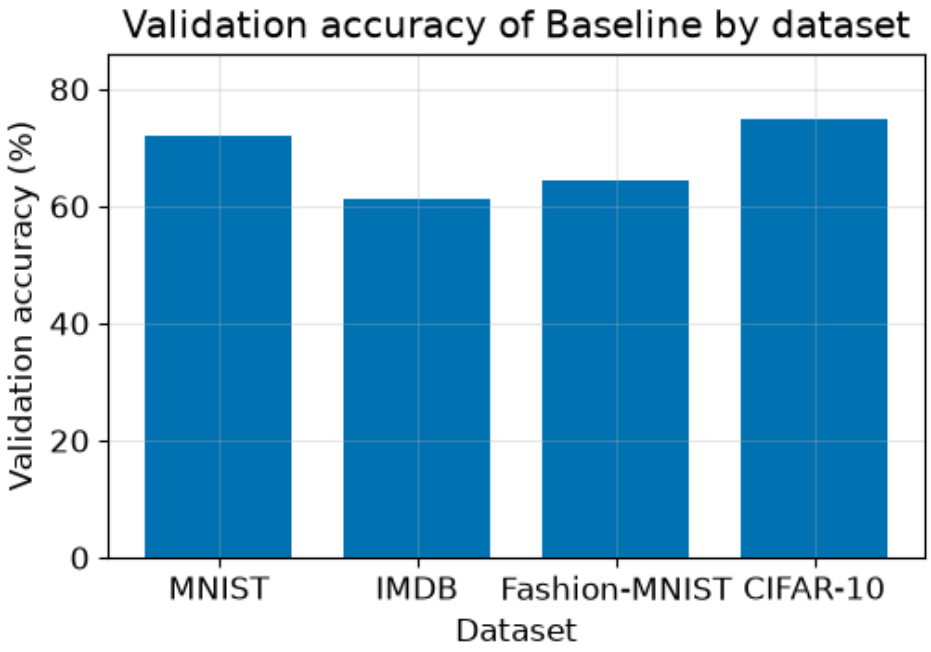


Figure 1: Validation accuracy of Baseline by dataset for the models in this experiment.

Baseline scores 10.6 points higher on MNIST than on IMDB (Figure 1). As shown in Figure 1, Baseline achieves 73.2% validation accuracy on CIFAR-10. On IMDB, Baseline obtains a validation accuracy of 69.9%. As shown in Figure 1, Baseline achieves about 60% validation accuracy on IMDB.

Figure 2 shows the f1 score by dataset.

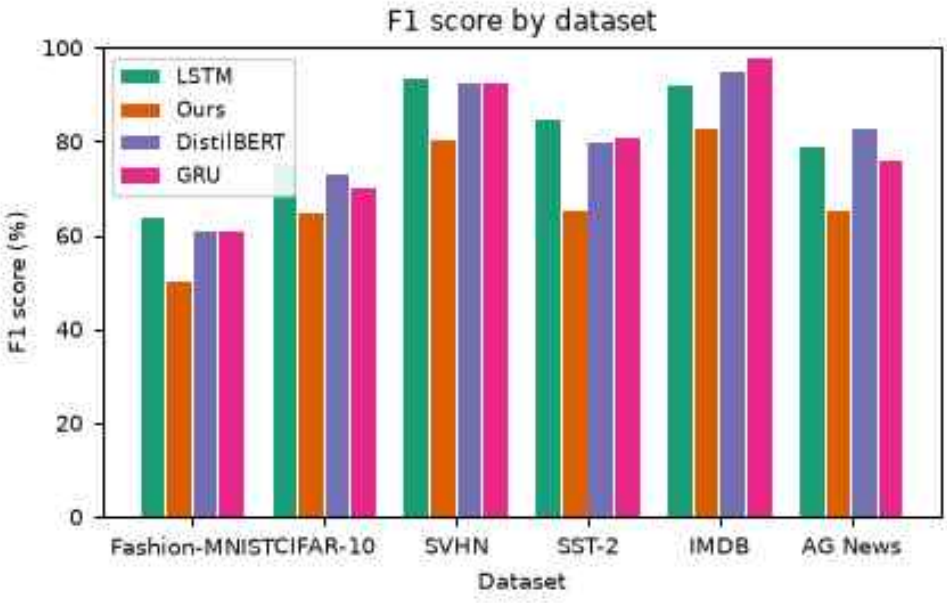


Figure 2: F1 score by dataset for the models in this experiment.

As shown in Figure 2, on SST-2, GRU obtains a F1 score of 80.7%. As shown in Figure 2, GRU outperforms DistilBERT by 4.8 percentage points on IMDB. On AG News, DistilBERT obtains a F1 score of 80.7%.

Figure 3 shows the f1 score by dataset.

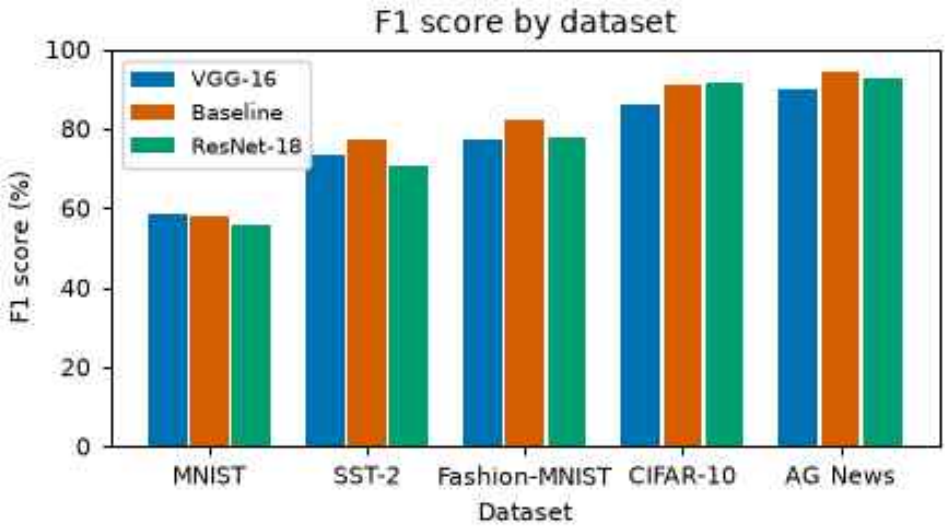


Figure 3: F1 score by dataset for the models in this experiment.

Baseline outperforms VGG-16 by 4 percentage points on SST-2 (Figure 3). Baseline achieves 99.3% F1 score on AG News. As shown in Figure 3, on Fashion-MNIST, VGG-16 obtains a F1 score of 77.6%. On Fashion-MNIST, VGG-16 obtains a F1 score of 77.3%.

3 Discussion

On IMDB, SVM obtains an accuracy of 53.5%. Overall, the results are in line with what we expected from the literature.